NovaMart — Python & SQL Quick Reference

1. Python / Pandas Quick Reference

Import

```
import pandas as pd
import numpy as np
```

Load CSV

```
df = pd.read_csv('file.csv')
```

Inspect

```
df.head(); df.shape; df.columns; df.info(); df.describe()
```

Missing values

```
df.isna().sum()
```

Duplicates

```
df.duplicated().sum()
```

Unique values

```
df['column'].nunique(); df['column'].unique()
```

Filter

```
df[df['Sales'] > 100000]
```

Multiple conditions

```
df[(df['Region']=='East') & (df['Profit'] > 100000)]
```

Group & aggregate

```
df.groupby('Category')['Sales'].sum()
```

Multiple aggregations

```
df.groupby('Category').agg(Sales=('Sales','sum'), Profit=('Profit','sum'))
```

Sort

```
df.sort_values('Sales', ascending=False)
```

Create calculated column

```
df['Profit_Margin'] = df['Profit'] / df['Sales'] * 100
```

Rename

```
df.rename(columns={'old':'new'}, inplace=True)
```

Drop duplicates

```
df.drop_duplicates(inplace=True)
```

Merge

```
df.merge(other, on='Customer_ID', how='left')
```

Date conversion

```
df['Order_Date'] = pd.to_datetime(df['Order_Date'])
```

Export

```
df.to_csv('output.csv', index=False)
```

2. SQL Quick Reference

SELECT

```
SELECT column1, column2 FROM table_name;
```

WHERE

```
SELECT * FROM orders WHERE Sales > 100000;
```

ORDER BY

```
SELECT * FROM orders ORDER BY Sales DESC;
```

GROUP BY

```
SELECT Category, SUM(Sales) FROM order_items GROUP BY Category;
```

HAVING

```
SELECT Category, SUM(Sales) FROM order_items GROUP BY Category HAVING SUM(Sales) >
1000000;
```

DISTINCT

```
SELECT DISTINCT Payment_Method FROM orders;
```

COUNT

```
SELECT COUNT(*) FROM orders;
```

CASE

```
SELECT CASE WHEN Profit > 0 THEN 'Profitable' ELSE 'Loss' END AS Status FROM
order_items;
```

INNER JOIN

```
SELECT * FROM customers c JOIN orders o ON c.Customer_ID = o.Customer_ID;
```

LEFT JOIN

```
SELECT * FROM customers c LEFT JOIN orders o ON c.Customer_ID = o.Customer_ID;
```

CTE

```
WITH sales AS (SELECT Category, SUM(Sales) AS total_sales FROM order_items GROUP
BY Category) SELECT * FROM sales;
```

Window function

```
SELECT Seller_ID, Sales, RANK() OVER (ORDER BY Sales DESC) AS rnk FROM
seller_sales;
```

3. Workplace Analysis Checklist

- Understand the business question
- Confirm data grain
- Validate IDs and relationships
- Profile missing/duplicate values
- Check dates and numerical ranges
- Define metrics clearly
- Analyze before visualizing
- Cross-check dashboard totals
- Document assumptions
- Communicate actionable insights